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Major Project Part -1

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Digital Twin for Live Condition Monitoring and Retrofit Decision Support of Reinforced Concrete Members

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Abstract

Flexural cracking reduces the stiffness of reinforced concrete beams and affects their serviceability, yet visual inspection alone cannot quantify this change or capture it between inspections. This project proposes a laboratory digital twin for live condition monitoring and retrofit decision support of these beams. Beams designed to IS 456, comprising a pilot, a damaged unstrengthened control and beams strengthened by concrete jacketing, bonded steel plates and fibre-reinforced polymer, will be tested in four-point bending, with load, deflection, acceleration and temperature acquired continuously. The twin will be evaluated on detecting loading events, lasting changes in response and sensor faults, flagging ambiguous evidence for inspection. Pilot tests will set damage stages below reinforcement yield. After each stage, a beam model will be updated by Bayesian inference to estimate flexural stiffness, and its prediction of the next service-load cycle will be recorded before that cycle is run. The twin will compare candidate strengthening schemes, designed to IS 15988 and ACI PRC-440.2-23, against a performance target for flexure, shear, bond and anchorage, and recommend a scheme for engineering review. Strengthened beams will be monitored from a new reference state, predicted service response will be compared with measurements, and failure loads will be assessed in destructive tests. The outcome is a tested prototype that shows when a beam has changed, by how much, and which repair restores the required performance.

Keywords: Digital Twin; Live Condition Monitoring; Reinforced Concrete Beams; Bayesian Model Updating; Retrofit Decision Support.